Generalization Limits of Protein Language Models

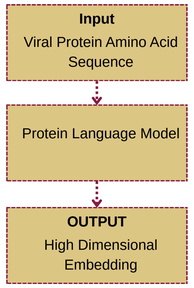
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Introduction

Problem

- Pre-trained protein language models (pLMs) create mathematical representations (embeddings) that capture the functional properties of proteins
- It is unknown whether these models actually learn meaningful biological rules or just memorize surface-level sequence similarities.



Goal

- Find whether these models can generalize when sequence similarity decreases by training a downstream model for virus host prediction at increasing taxonomic train/test splits

Methodology

Downstream Model



- Trained a multi-class virus host prediction model on top of three state-of-the-art protein language models (HAVEN[1], ESM2[2], LucaVirus[3])
- Downstream model was 2-Layer Multi-Layer Perceptron (MLP) and was tasked with predicting host species.
- Viral proteins in dataset were **randomly** split, split by **species**, **genus**, and **family** restricting sequence identity between train and test sets.
- Each split for each model was run for 5 iterations with different seeds

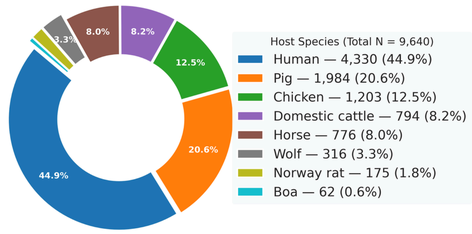
Evaluation

- Model performance evaluated using Area Under The Precision Recall Curve (AUPRC)
- Basic Local Alignment Search Tool (BLAST), was also run on all splits as a completely sequence-based baseline to compare model performance to.

Hyperparameters

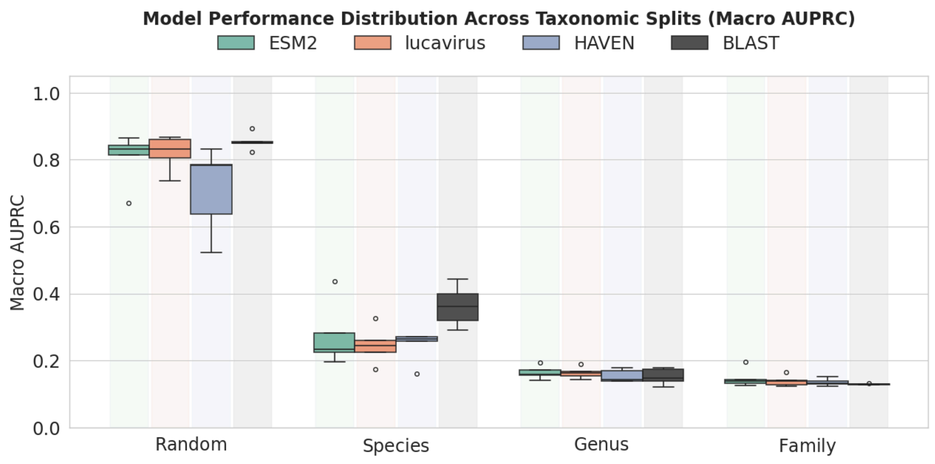
- Downstream MLP models were trained for 25 epochs. Focal loss with gamma = 2 to handle class imbalance. Hyperparameters were tuned, yielding optimal batch size of 32, learning rate of 1e-3, and weight decay of 1e-4.

Data

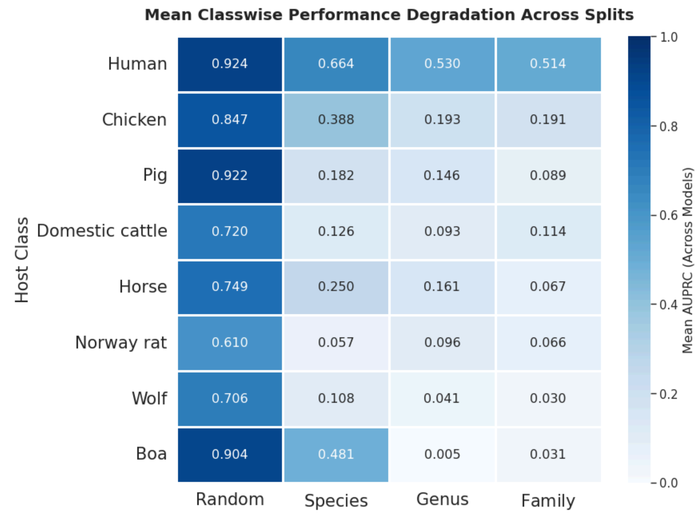


- 10,000 viral proteins from the BVBRC database.
- 8 multi-class host species labels
- Clustered by sequence similarity to minimize duplicate bias.

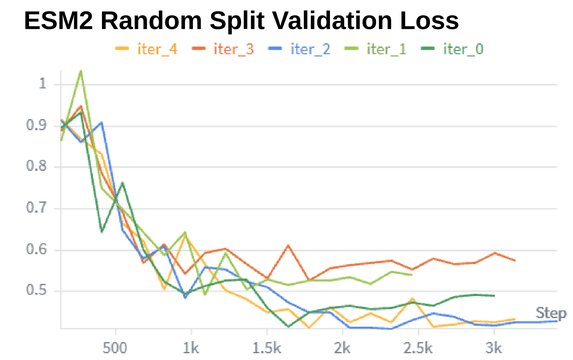
Results

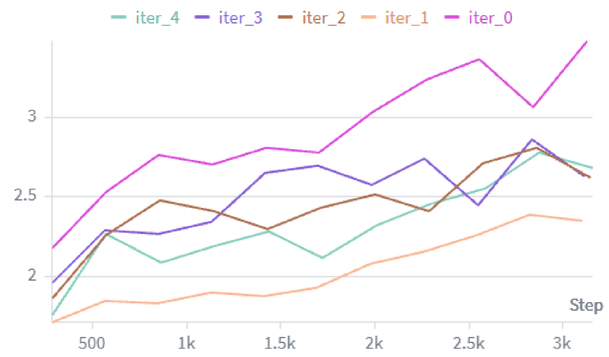


- Macro AUPRC for each model decreased as taxonomic split increased
- Macro AUPRC of BLAST decreased at a similar rate as the models



- Mean class wise AUPRC averaged across all models
- Every class showed decrease in AUPRC as taxonomic split increased





- On random split, validation loss smoothly decreased while on family split, it never converges

Summary

- Performance drop at increasingly broader taxonomic levels suggests that downstream models using pLM embeddings are highly susceptible to sequence-level memorization and struggle to generalize to unseen viral lineages. Alignment between pLM classifier decay and BLAST performance further shows that embeddings from pLM's may only show surface level sequence similarities.
- Future work should explore training objectives that explicitly penalize sequence memorization.

References & Acknowledgments

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